Data-Driven Follow-Up Assurance for Rural Teleconsultations

Problem Statement

Teleconsultation is now the primary link between rural patients and doctors in Telangana. At Ayushman Arogya Mandirs, a Community Health Officer connects patients with hypertension and diabetes to district hospital doctors over eSanjeevani. However, the consultation is only the first step: patients must collect medicines, complete tests and return for review within weeks. A large share never do, and the system records the consultation as "completed" with no visibility of what happened afterwards. Follow-up data sits scattered across teleconsultation logs, NCD screening records, dispensing registers and ASHA diaries, and is never linked.

Your task

Design a data-driven follow-up assurance solution that can combine teleconsultation records, NCD screening data, medicine dispensing and visit registers, patient demographics and geographic data to identify patients lost to follow-up, predict who is at risk of dropping out, and guide health workers to bring them back into care. The solution should demonstrate how data linkage and AI can help rural health systems move from counting consultations to ensuring completed care within the constraints of low-end phones, intermittent connectivity, and limited health-worker time.

Teams should address

- How can fragmented teleconsultation, NCD, dispensing and visit records be linked to determine a patient's follow-up status?
- What stages and factors contribute to patients being "lost to follow-up", and how can these patients be identified?
- What data-driven intervention should be implemented to improve follow-up, and how would its effectiveness, feasibility and cost be measured?

Dataset

A synthetic dataset representing the fragmented healthcare records described in the case will be provided to the teams. The dataset contains information from multiple stages of the patient care journey.

Teams are expected to use the provided dataset to perform data-driven analysis and support their proposed solution with relevant findings and evidence.

Constraints and Considerations

Solutions should consider privacy-preserving, ABDM-compliant data handling and consent, as well as the practical constraints of rural healthcare delivery, including low-end phones, intermittent connectivity and limited health-worker time.